# Decision Trees for Classification and Regression

*Investigating decision tree behaviour and hyperparameter effects on Iris, Diabetes, and synthetic datasets using scikit-learn.*

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## Overview

Decision trees are interpretable models used for both classification and regression. This project applies them to three datasets: the Iris dataset (150 samples, 4 features, 3 classes) for classification, the Diabetes dataset for regression, and a synthetic noisy sine wave. Seven topics are explored systematically, from basic data statistics through hyperparameter tuning and decision surface visualisation.

## Goals

Develop a thorough understanding of how decision tree hyperparameters affect model fit, and identify the settings that best balance training accuracy and generalisation on diverse datasets.

## Objectives

- Summarise Iris feature statistics and class balance as a baseline for understanding the data.
- Quantify how max_depth and min_samples_leaf jointly affect classification accuracy on Iris.
- Analyse the effect of training on only 5 percent of the data and recover performance via GridSearchCV.
- Visualise decision boundaries for every pair of Iris features to identify the most discriminative attributes.
- Fit regression trees on both a noisy sine wave and the Diabetes dataset, then tune max_depth to minimise test RMSE.

## Approach

Each topic is implemented end-to-end in a single notebook. Classification experiments use a 70/30 train/test split on Iris; regression experiments use the Diabetes dataset with the default sklearn split. Grid search is applied with cross-validation to select optimal hyperparameters before evaluating on held-out data. Decision surfaces are plotted for each feature pair to give a geometric view of the learned boundaries.

## Outcomes

- Petal length and petal width were identified as the most discriminative Iris features, producing near-clean three-class separation.
- Shallow trees (max_depth of 2 to 3) matched the performance of deeper trees on Iris while avoiding overfitting.
- With only 5 percent of training data, grid search selected a shallow tree that substantially reduced misclassification compared to the default.
- On the Diabetes regression task, a constrained tree found by grid search outperformed an unbounded tree, confirming that depth control is essential on noisy, low-signal data.

- Across all experiments, controlling tree depth via max_depth and min_samples_leaf was the single most effective lever for balancing bias and variance.